Sequential Depletion Ordering with Residual-Fraction Costs

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Abstract

A fixed collection of positive loads is removed sequentially from a resource with permanent reserve $q \geq 0$. If removing size x at residual level R costs the residual fraction x/R, the cost of a permutation π is

$$C_q(\pi) = \sum_{k=1}^{n} \frac{x_{\pi(k)}}{q + \sum_{j=k}^{n} x_{\pi(j)}}.$$

An exact adjacent-interchange identity shows that nondecreasing size order minimizes C_q and nonincreasing order maximizes it. For $q > 0$, the log-depletion vector $d_k = \log(R_k/R_{k+1})$ is majorized between the decreasing-size and increasing-size extremes, so every Schur-convex functional of d shares those extremes. The same comparison classifies nonlinear stage costs $\sum \phi(x_{\pi(k)}/R_k)$ by the curvature of $g(t) = \phi(1 - e^{-t})$.

With precedence constraints, an order-ideal dynamic program computes the optimum and smallest-available greedy can fail. For independent sizes totally ordered by monotone likelihood ratio, the ex-ante small-first schedule minimizes expected cost. For multi-resource loads, componentwise chains retain the sorting rule, while heterogeneous vectors admit residual-state preference reversals. Related residual-fraction sums appear in cumulative Riemann-sum bounds; those results give order-independent integral estimates rather than a permutation theory. The contribution is the ordering, majorization, constrained, stochastic, and network analysis built on one interchange identity.

Keywords: Scheduling, residual fraction, adjacent interchange, majorization, stochastic orders, precedence constraints

1 Introduction

Suppose a system has residual size R and a load of size x is removed. The relative shock is x/R. If several fixed loads must be removed in some order, each removal changes the residual, so the sequence of shocks depends on the order.

Let $x_1, \ldots, x_n > 0$ and let $q \geq 0$ be reserve that is never removed. For a permutation π of $[n] = \{1, \ldots, n\}$, define

$$C_q(\pi) = \sum_{k=1}^{n} \frac{x_{\pi(k)}}{q + \sum_{j=k}^{n} x_{\pi(j)}}. \tag{1}$$

General sum-of-ratios programs are hard (Schaible and Shi 2003). The suffix structure in (1) is special: an exact adjacent-interchange identity determines the optimum.

The main facts are these.

1. Nondecreasing size order minimizes C_q; nonincreasing order maximizes it.
2. For $q > 0$, the log-depletion vectors of all permutations lie between the decreasing-size and increasing-size extremes in the majorization order.
3. Stage costs $\phi(x/R)$ inherit those extremal orders according to the curvature of $g(t) = \phi(1 - e^{-t})$.

4. With precedence constraints, the optimum is given by a dynamic program over order ideals; the natural smallest-available rule is not always optimal.
5. If independent random sizes are totally ordered by monotone likelihood ratio, the corresponding fixed order minimizes expected cost.
6. Componentwise-comparable multi-resource loads still sort; for heterogeneous loads, even the locally preferred order can depend on the residual state.

The deterministic swap argument is classical adjacent interchange in the sense of Smith (1956), Sidney (1975), and Lawler (1978). Closely related residual-ratio sums also occur as cumulative Riemann sums (Pain 2026). For $q > 0$, reverse normalization identifies (1) exactly with one such sum and yields an order-independent integral bound. That result does not compare permutations. The contribution of this paper is the permutation theory for (1) and the majorization, constrained, stochastic, and multi-resource results built on its interchange identity.

2 Related work

2.1 Adjacent interchange and scheduling

Local pairwise comparisons are standard in single-machine sequencing (Smith 1956; Sidney 1975; Lawler 1978; Correa and Schulz 2005; Jäger and Warode 2024). Those arguments usually target completion-time objectives. The cost (1) is a nested residual-fraction sum, not weighted completion time.

2.2 Cumulative residual fractions

Pain (2026) studies sums $\sum_{i=1}^{n} a_i g(S_i)$ with $a_i > 0$, $\sum_i a_i = 1$, and prefix totals $S_i = \sum_{j=1}^{i} a_j$. For decreasing integrable g, that work proves

$$\sum_{i=1}^{n} a_i g(S_i) \leq \int_0^1 g(x)\,dx,$$

an order-independent integral upper bound. To make the connection exact, let $X = \sum_i x_i$, set $\rho = q/X$, reverse the schedule via $\sigma(i) = \pi(n+1-i)$, and define

$$a_i = \frac{x_{\sigma(i)}}{X}, \qquad S_i = \sum_{j=1}^{i} a_j.$$

Then, for $q > 0$,

$$C_q(\pi) = \sum_{i=1}^{n} \frac{a_i}{\rho + S_i} \leq \int_0^1 \frac{ds}{\rho + s} = \log\left(1 + \frac{X}{q}\right). \tag{2}$$

The inequality is Pain's theorem with $g(s) = 1/(\rho + s)$. It controls every order but neither compares two orders nor identifies an extremal permutation. The case $q = 0$ has the same cumulative form with $g(s) = 1/s$, but that function is not integrable at zero and is outside the stated integral-bound theorem.

2.3 Majorization

Majorization and Karamata's inequality are classical (Hardy et al. 1952; Marshall et al. 2011; Arnold 2007). Here they apply to the vector of log residual ratios generated by a permutation.

2.4 Sequential choice and size-biased order

Stagewise terms $x_i / \sum_{j \in S} x_j$ appear as choice probabilities in Luce and Plackett–Luce models and in size-biased residual allocation (Luce 1959; Plackett 1975; Perman et al. 1992; Pitman and Tran 2015). That work studies random orders and products of choice probabilities rather than optimization of the deterministic sum (1).

2.5 Stochastic orders and rearrangement

Monotone likelihood ratio is a standard stochastic order (Karlin and Rubin 1956; Shaked and Shanthikumar 2007). Bivariate characterizations and stochastic rearrangement inequalities provide general foundations for pairwise interchange under likelihood-ratio order (Shanthikumar and Yao 1991; Chang and Yao 1993). This paper applies that established method to the explicit swap kernel of C_q.

2.6 Precedence and order ideals

Dynamic programming over order ideals is a standard exact method for separable costs on linear extensions of a poset (Steiner 1984). The recursion below specializes that classical state space to residual-fraction stage costs, with a width-based complexity bound and a greedy counterexample for this objective.

3 The deterministic depletion objective

For a permutation π, write

$$R_k(\pi) = q + \sum_{j=k}^{n} x_{\pi(j)}$$

for the resource present immediately before stage k. Because every size is positive, $R_k(\pi) \geq x_{\pi(k)} > 0$ for all k, including the case $q = 0$.

Lemma 3.1 (Exact adjacent-interchange identity) *Suppose two adjacent items have sizes $a, b > 0$, and let $T \geq 0$ be the reserve plus the total size of all items scheduled after them. Then*

$$\begin{aligned} &C_q(\dots, a, b, \dots) - C_q(\dots, b, a, \dots) \\ &\quad = \frac{ab(a-b)}{(T+a+b)(T+a)(T+b)}. \end{aligned} \tag{3}$$

Proof Only the two displayed stages change. Their difference is

$$\begin{aligned} &\left(\frac{a}{T+a+b} + \frac{b}{T+b}\right) - \left(\frac{b}{T+a+b} + \frac{a}{T+a}\right) \\ &= \frac{a-b}{T+a+b} + \frac{T(b-a)}{(T+a)(T+b)} \\ &= \frac{ab(a-b)}{(T+a+b)(T+a)(T+b)}. \end{aligned}$$

All denominators are positive because $a, b > 0$ and $T \geq 0$. $\square$

Theorem 3.2 (Extremal orders) *For every $q \geq 0$, nondecreasing size order minimizes C_q, and nonincreasing size order maximizes C_q. The minimizer and maximizer are unique up to permutations of equal-sized items.*

Proof If an adjacent pair is inverted, $a > b$, then lemma 3.1 shows that swapping it from (a, b) to (b, a) strictly decreases the objective. Repeatedly removing inversions yields nondecreasing order. Reversing the argument yields nonincreasing order as the maximizer. $\square$

When $q = 0$, the same theorem gives a harmonic bound.

Corollary 3.3 (Harmonic sandwich) *Let $H_n = \sum_{k=1}^{n} 1/k$. For $q = 0$,*

$$C_0(x^{\uparrow}) \leq H_n \leq C_0(x^{\downarrow}),$$

with equality on either side if and only if all sizes are equal.

Proof For $y_1 \leq \dots \leq y_n$,

$$\sum_{j=k}^{n} y_j \geq (n-k+1)y_k,$$

so the kth term is at most $1/(n-k+1)$. The descending inequality is analogous. $\square$

4 Log-depletion majorization

Assume throughout this section that $q > 0$, so every residual is strictly larger than the load about to be removed and every log increment is finite. Define the fractional shock and log-depletion shock at stage k by

$$\begin{aligned} z_k(\pi) &= \frac{x_{\pi(k)}}{R_k(\pi)}, \\ d_k(\pi) &= \log \frac{R_k(\pi)}{R_{k+1}(\pi)} = -\log(1 - z_k(\pi)). \end{aligned} \tag{4}$$

The total log depletion telescopes:

$$\sum_{k=1}^{n} d_k(\pi) = \log \frac{q + \sum_i x_i}{q}, \tag{5}$$

independently of π.

For vectors $u, v \in \mathbb{R}^n$ with equal coordinate sums, write $u \prec v$ when u is majorized by v; equivalently, the sums of the m largest coordinates of u do not exceed those of v for $m < n$ (Marshall et al. 2011).

Theorem 4.1 (Log-depletion majorization sandwich) *Let $d(\pi) = (d_1(\pi), \dots, d_n(\pi))$. Then every permutation satisfies*

$$d(x^{\downarrow}) \prec d(\pi) \prec d(x^{\uparrow}). \tag{6}$$

Thus decreasing size order produces the least concentrated log-depletion vector, and increasing size order produces the most concentrated one.

Proof Consider an adjacent inversion $a \geq b$ with tail resource $T > 0$. Before swapping, the two log shocks are

$$u_1 = \log \frac{T+a+b}{T+b}, \qquad u_2 = \log \frac{T+b}{T}.$$

After swapping to (b, a), they are

$$v_1 = \log \frac{T+a+b}{T+a}, \qquad v_2 = \log \frac{T+a}{T}.$$

The pair sums are equal. Also

$$\begin{aligned} u_2 &\leq v_2, \\ u_1 &= \log\left(1 + \frac{a}{T+b}\right) \\ &\leq \log\left(1 + \frac{a}{T}\right) = v_2. \end{aligned}$$

Hence $\max\{u_1, u_2\} \leq \max\{v_1, v_2\}$. For two-dimensional vectors of equal sum, this is exactly $(u_1, u_2) \prec (v_1, v_2)$. Replacing two coordinates by a vector that majorizes them preserves majorization after adjoining all unchanged coordinates. Repeated swaps from decreasing to increasing size order prove (6). □

Corollary 4.2 (Universal convex-risk ordering) *For every convex function g on $[0, \infty)$,*

$$\sum_k g(d_k(x^{\downarrow})) \leq \sum_k g(d_k(\pi)) \leq \sum_k g(d_k(x^{\uparrow})).$$

For concave g, both inequalities reverse.

Proof Apply Karamata's inequality to theorem 4.1 (Hardy et al. 1952; Marshall et al. 2011). □

Consequently, decreasing size order minimizes the maximum log shock, every ℓ_p norm with $p > 1$, the variance of the log-shock vector, and every separable convex risk measure on d. Increasing order minimizes the raw fraction sum because $z = 1 - e^{-d}$ is concave in d.

4.1 Nonlinear stage penalties and a phase transition

Let $\phi : [0, 1) \to \mathbb{R}$ and define

$$J_{\phi,q}(\pi) = \sum_{k=1}^{n} \phi(z_k(\pi)). \tag{7}$$

Set

$$g_\phi(t) = \phi(1 - e^{-t}).$$

Corollary 4.3 (Curvature classification) *If g_ϕ is concave, increasing size order minimizes $J_{\phi,q}$ and decreasing size order maximizes it. If g_ϕ is convex, decreasing size order minimizes $J_{\phi,q}$ and increasing size order maximizes it.*

If ϕ is twice differentiable and $z = 1 - e^{-t}$, then

$$g''_\phi(t) = (1-z)\big((1-z)\phi''(z) - \phi'(z)\big). \tag{8}$$

The order-invariant boundary solves

$$(1-z)\phi''(z) = \phi'(z),$$

so

$$\phi(z) = A[-\log(1-z)] + B. \tag{9}$$

An explicit family makes the transition concrete. For $\alpha \in \mathbb{R}$, define

$$\phi_\alpha(z) = \begin{cases} \dfrac{(1-z)^{-\alpha} - 1}{\alpha}, & \alpha \neq 0, \\ -\log(1-z), & \alpha = 0. \end{cases} \tag{10}$$

Then

$$\phi_\alpha(1 - e^{-t}) = \begin{cases} (e^{\alpha t} - 1)/\alpha, & \alpha \neq 0, \\ t, & \alpha = 0, \end{cases}$$

and its second derivative has the sign of α.

Corollary 4.4 (Small-first / log-invariant / large-first transition) *For the family (10):*

(i) if $\alpha < 0$, increasing size order minimizes $J_{\phi_\alpha,q}$;

(ii) if $\alpha = 0$, every order has identical cost;

(iii) if $\alpha > 0$, decreasing size order minimizes $J_{\phi_\alpha,q}$.

In particular, $\alpha = -1$ recovers the raw fraction $\phi(z) = z$, while $\alpha = 1$ recovers the post-removal odds $\phi(z) = z/(1-z)$.

Three natural ways to score depletion therefore sit on opposite sides of the same line: residual share before removal, pure log change, and residual share after removal.

5 Precedence-constrained depletion

Let $P = (V, \preceq)$ be a finite precedence poset. A feasible schedule is a linear extension of P. For $S \subseteq V$, let

$$Q(S) = q + \sum_{j \in V \setminus S} x_j.$$

A set S is an order ideal if $j \in S$ and $i \preceq j$ imply $i \in S$. Let $A(S)$ be the set of minimal elements of $V \setminus S$; these are the jobs available next. As in the unconstrained case, every size is positive, so $Q(S) > 0$ for every proper ideal S.

Theorem 5.1 (Exact dynamic program over ideals) *Define $F(V) = 0$ and, for every proper order ideal S,*

$$F(S) = \min_{i \in A(S)} \left\{ \frac{x_i}{Q(S)} + F(S \cup \{i\}) \right\}. \tag{11}$$

Then $F(\varnothing)$ is the minimum of C_q over all linear extensions of P.

Proof Every feasible continuation from S must choose an available minimal element $i \in A(S)$. Its immediate denominator is $Q(S)$, independent of which available job is selected. After choosing i, the processed set is the ideal $S \cup \{i\}$. The principle of optimality gives (11). □

Let $\mathcal{I}(P)$ be the set of order ideals and let w be the width of P.

Proposition 5.2 (Width-sensitive complexity) *With available sets maintained incrementally, (11) can be evaluated in $O(w|\mathcal{I}(P)|)$ state transitions, up to input-preprocessing costs. If a width-w chain decomposition has chain lengths $n_1, \ldots, n_w$, then*

$$|\mathcal{I}(P)| \le \prod_{r=1}^{w} (n_r + 1), \tag{12}$$

so the algorithm is polynomial for every fixed width w.

Proof The available set is an antichain, hence has size at most w. An ideal intersects each chain in a prefix, and therefore injects into the tuple of its w prefix lengths, proving (12). □

In general this is exponential in n. For fixed width it is $n^{O(w)}$. The unconstrained increasing-order cost of the remaining jobs is also an admissible lower bound for branch-and-bound.

Lemma 5.3 (Local inversion exclusion) *No optimal linear extension contains adjacent incomparable jobs i, j with $x_i > x_j$ in that order.*

Proof Because the jobs are incomparable and adjacent, swapping them preserves feasibility. The swap strictly reduces cost by lemma 3.1. □

Local inversion exclusion does not justify a smallest-available greedy policy.

Example 5.4 (Smallest available can be suboptimal) Let $q = 1$ and let $A \prec C$, with B incomparable to both. Set

$$(x_A, x_B, x_C) = (8, 7, 2).$$

Initially, A and B are available, so smallest-available chooses B, producing

$$C_1(B, A, C) = \frac{7}{18} + \frac{8}{11} + \frac{2}{3} \approx 1.7828.$$

The feasible order (A, C, B) has

$$C_1(A, C, B) = \frac{8}{18} + \frac{2}{10} + \frac{7}{8} \approx 1.5194.$$

Processing the larger predecessor first unlocks the very small successor and is globally preferable.

6 Stochastic sizes under likelihood-ratio order

Suppose sizes are independent positive random variables $X_1, \ldots, X_n$, and the schedule must be selected ex ante. Let all variables admit densities with respect to a common dominating measure. Recall that $X \le_{\mathrm{lr}} Y$ if $f_Y(x)/f_X(x)$ is nondecreasing wherever defined (Shaked and Shanthikumar 2007).

Theorem 6.1 (Stochastic small-first rule under MLR order) *Assume*

$$X_1 \le_{\mathrm{lr}} X_2 \le_{\mathrm{lr}} \cdots \le_{\mathrm{lr}} X_n.$$

Then the order $(1, 2, \ldots, n)$ minimizes $\mathbb{E}[C_q(\pi)]$ over all fixed permutations π. Reversing the order maximizes the expectation. Under strict likelihood-ratio inequalities on sets of positive measure, the extremal orders are unique up to identically distributed jobs.

Proof It suffices to compare an adjacent pair X, Y with $X \leq_{\mathrm{lr}} Y$. Let $T \geq 0$ be the reserve plus the sum of all random sizes after the pair. Independence makes T independent of (X, Y). Conditional on $T = t$, lemma 3.1 gives

$$h_t(x, y) = \frac{xy(x-y)}{(t+x+y)(t+x)(t+y)} \tag{13}$$

after subtracting the cost of order (Y, X) from that of order (X, Y). The kernel is antisymmetric and positive when $x > y$. Pairing the regions $x > y$ and $y > x$ yields

$$\mathbb{E}[h_t(X, Y)] = \int_{x>y} h_t(x, y) \times \left[f_X(x) f_Y(y) - f_X(y) f_Y(x)\right] dx\, dy.$$

For $x > y$, likelihood-ratio order implies

$$\frac{f_Y(x)}{f_X(x)} \geq \frac{f_Y(y)}{f_X(y)},$$

so the bracketed factor is nonpositive. Therefore $\mathbb{E}[h_t(X, Y)] \leq 0$ for every t, and hence

$$\mathbb{E}[C_q(\ldots, X, Y, \ldots)] \leq \mathbb{E}[C_q(\ldots, Y, X, \ldots)].$$

Adjacent interchanges sort the jobs into likelihood-ratio order. Reversing each comparison gives the maximizer. □

Corollary 6.2 (Common parametric families) *The stochastic theorem applies to, among others:*

(i) exponential sizes, ordered by decreasing rate;

(ii) gamma sizes with common shape, ordered by decreasing rate;

(iii) lognormal sizes with common variance, ordered by increasing log-location parameter.

Ordering by expected size is not enough. Likelihood-ratio order controls the sign of the full antisymmetric swap kernel in every tail state.

7 Multi-resource and network depletion

Let there be m resource pools. Item i has a nonnegative load vector

$$x_i = (x_{i1}, \ldots, x_{im}) \in \mathbb{R}_+^m$$

that is not the zero vector, resource r has reserve $q_r > 0$, and $w_r > 0$ is its importance weight. The network objective is

$$C^{\mathrm{net}}(\pi) = \sum_{r=1}^{m} w_r \sum_{k=1}^{n} \frac{x_{\pi(k),r}}{q_r + \sum_{j=k}^{n} x_{\pi(j),r}}, \tag{14}$$

The positive reserves keep every resourcewise denominator well defined even when an item has zero load on a particular resource.

Lemma 7.1 (Network swap identity) *For adjacent load vectors $a, b \in \mathbb{R}_+^m$ and tail resources $T_r > 0$,*

$$\Delta(a, b; T) = \sum_{r=1}^{m} w_r \frac{a_r b_r (a_r - b_r)}{(T_r + a_r + b_r)(T_r + a_r)(T_r + b_r)}, \tag{15}$$

where Δ is the cost of (a, b) minus the cost of (b, a).

Proof The network objective is a nonnegative weighted sum of single-resource objectives. The algebra in lemma 3.1 remains valid when $a_r = 0$ or $b_r = 0$ because $T_r > 0$. Summing the weighted resourcewise differences gives (15). □

Theorem 7.2 (Componentwise-chain ordering) *Suppose the item load vectors form a chain under componentwise order. Then any componentwise nondecreasing order minimizes C^{net}, and the reverse order maximizes it. In particular, if $x_i = s_i v$ for a common profile $v \in \mathbb{R}_+^m \setminus \{0\}$ and scalars $s_i > 0$, sorting the scalar magnitudes s_i increasingly is optimal.*

Proof If $a \geq b$ componentwise, every summand in (15) is nonnegative. Swapping (a, b) to (b, a) therefore cannot increase cost. Repeated adjacent interchanges prove the claim. □

For resources with $q_r > 0$, each resource has its own log-depletion vector. Under the componentwise-chain assumption, the same adjacent swaps move every such resource's log vector in the same majorization direction. Decreasing componentwise order therefore minimizes every separable convex risk

$$\sum_{r=1}^{m} w_r \sum_{k=1}^{n} g_r(d_{k,r})$$

with convex g_r, while increasing order minimizes the raw-fraction objective (14).

The chain assumption is essentially the limit of any context-free sorting rule.

Theorem 7.3 (Context reversal for pairwise priorities) *For heterogeneous load vectors, the locally preferred order of a fixed pair can reverse as the residual network state changes. Consequently, no priority relation depending only on the two item vectors can agree with the optimal adjacent order for all tail states.*

Proof Take two resources with equal weights and

$$a = (4, 1), \qquad b = (1, 3).$$

For tail state $T = (1, 1)$, (15) gives

$$\Delta(a, b; (1, 1)) = \frac{4 \cdot 1 \cdot 3}{6 \cdot 5 \cdot 2} + \frac{1 \cdot 3 \cdot (-2)}{5 \cdot 2 \cdot 4} = \frac{1}{20} > 0.$$

Hence (b, a) is locally cheaper. For tail state $T = (10, 1)$,

$$\Delta(a, b; (10, 1)) = \frac{2}{385} - \frac{3}{20} = -\frac{223}{1540} < 0,$$

so (a, b) is locally cheaper. Since the same pair requires opposite orders in two residual states, no context-free relation on item vectors alone can be universally correct. □

This does not rule out state-dependent policies, dynamic programming, approximation algorithms, or static indices on restricted load geometries. It shows that no scalar priority can reproduce the correct adjacent comparison in every residual state of the unrestricted network model.

8 Economic and computing interpretations

The model applies only when stage cost is genuinely proportional to the share of capacity present immediately before the action. It is not a substitute for travel cost, fixed setup cost, deadline penalties, price impact, or other domain-specific effects.

8.1 Computing systems

Natural targets include staged retirement of servers, shards, replicas, or service capacity; memory reclamation; and dependency-constrained decommissioning. The raw-fraction objective favors removing small components first. Convex functions of log depletion favor large-first schedules that smooth multiplicative shocks. In a multi-resource system, loads may represent CPU, memory, bandwidth, storage, or regional redundancy. Theorem 7.3 says heterogeneous systems generally need state-aware scheduling.

8.2 Logistics and infrastructure

Loads may represent warehouse zones, generation units, supplier capacity, or temporary emergency resources. Precedence captures operational dependencies. The stochastic theorem applies when uncertain loads sit in a likelihood-ratio-ordered family, which is stronger than sorting means.

8.3 Economics

In a stylized market-exit or subsidy-withdrawal model, x_i/R_k is a participant's share of the remaining system. Increasing order minimizes the sum of ordinary percentage shocks. Decreasing order minimizes convex concentration risk in continuously compounded shocks. A reserve q can represent outside liquidity, a backstop, permanent capacity, or a diversified outside option. Larger q weakens every pairwise ordering penalty in (3).

Corollary 4.4 is the policy point: "minimize disruption" is incomplete until disruption is specified as a concave aggregate of ordinary shares, an additive log change, or a convex tail-risk measure.

9 Limitations and open problems

1. **Complexity under precedence.** There is an exact ideal DP and a greedy counterexample, but no NP-hardness classification or approximation guarantee.
2. **Stochastic orders.** Likelihood-ratio order is strong. Whether hazard-rate, dispersive, or usual stochastic order suffices for useful subclasses is open.
3. **Network case.** Context reversal rules out a universal context-free adjacent priority rule. Complexity and approximability of the unrestricted multi-resource problem remain open.
4. **Empirical calibration.** Applications need evidence that fraction or log shock is the actual cost, not just a convenient surrogate.

5. **Online models.** Items may arrive over time or reveal size only when processed. Competitive analysis and regret bounds are natural next steps.

10 Conclusion

A nested residual-fraction objective has an exact adjacent-interchange law: small-first minimizes the raw cost, large-first maximizes it. The log representation gives a stronger majorization fact: large-first smooths multiplicative depletion, while small-first concentrates it. The same identity drives an order-ideal DP under precedence, an MLR expectation theorem under independent random sizes, and a componentwise sorting rule for multi-resource chains. In heterogeneous networks, even a fixed pair can reverse local preference with the residual state, marking a boundary for context-free pairwise priorities.

Related cumulative residual-fraction sums already admit order-independent integral bounds (Pain 2026). The claim of this research is the permutation theory for (1) and the package around one interchange identity, not a new residual-ratio expression in isolation and not a new interchange technique.

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Competing interests

The author declares no competing interests.

Data availability

No datasets were generated or analysed in this study.

Code availability

Companion Python and Kotlin verification scripts supplied with the manuscript perform randomized numerical checks of the swap identity and majorization theorem, enumerate the precedence counterexample, evaluate the two network context states, and Monte Carlo check the stochastic theorem for exponential sizes. The scripts are provided for transcription checks only and do not replace the proofs.

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